DSAIT4110 Assignment 2: Model Predictive Control (MPC) Analysis Report

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1 Introduction

This report summarizes the simulation results for the three MPC modules:

- Module 1: prediction and system behavior (open-loop vs closed-loop intuition),
- Module 2: constrained MPC behavior under horizon and cost tuning,
- Module 3: Tube MPC intuition (nominal vs actual trajectories under bounded disturbances).

All figures were generated from the Python scripts in `mpc/` and saved under `mpc/figures/`.

2 Module 1: Prediction and System Behavior

2.1 Base Open-Loop Run

Figure **??** shows the open-loop trajectories for the default setup

$$x_0 = [-2,\ 3]^\top, \quad u = 0.1, \quad d = [0,\ 0]^\top.$$

The state x_2 grows approximately linearly, while x_1 grows faster due to accumulation through the coupling term in A:

$$x_{1,k+1} = x_{1,k} + 0.1\, x_{2,k}.$$

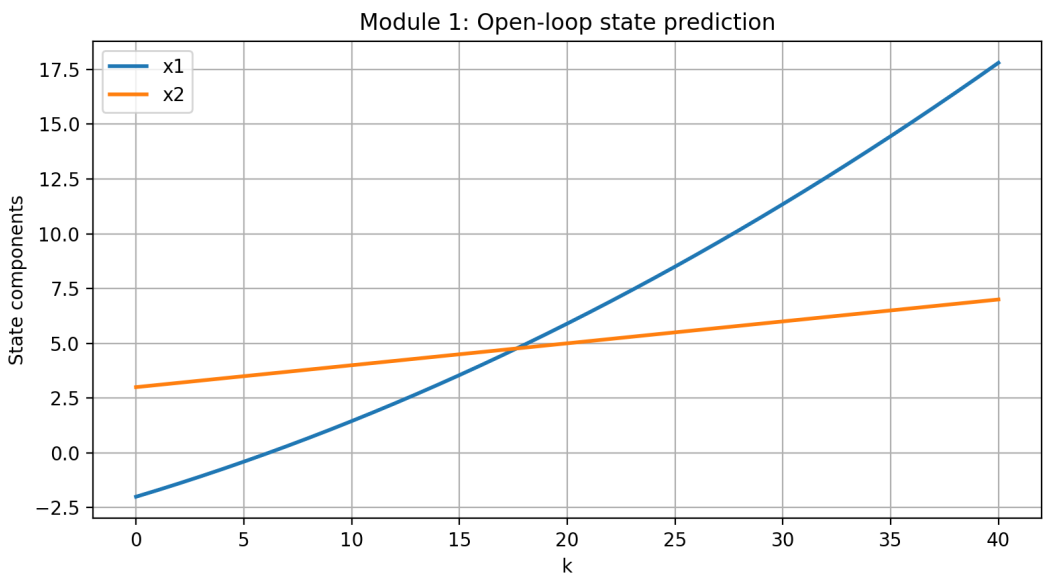

Figure 1: Module 1 open-loop prediction (base run).

2.2 Analysis

The plot confirms that open-loop behavior is sensitive to initial conditions and model dynamics, since no feedback correction is applied online. In this run, positive x_2 keeps integrating into x_1, which explains why x_1 eventually dominates in magnitude. This supports the motivation for closed-loop control: without feedback, disturbances or state offsets persist and can grow.

3 Module 2: MPC and Horizon Effects

3.1 Base Run ($N_p = 3$)

Figure **??** shows the constrained MPC closed-loop behavior for

$$N_p = 3, \quad Q = \mathrm{diag}(5, 1), \quad R = 0.1, \quad u \in [-0.8, 0.8].$$

Both states converge smoothly toward the origin. The applied control remains small and does not hit the input bounds.

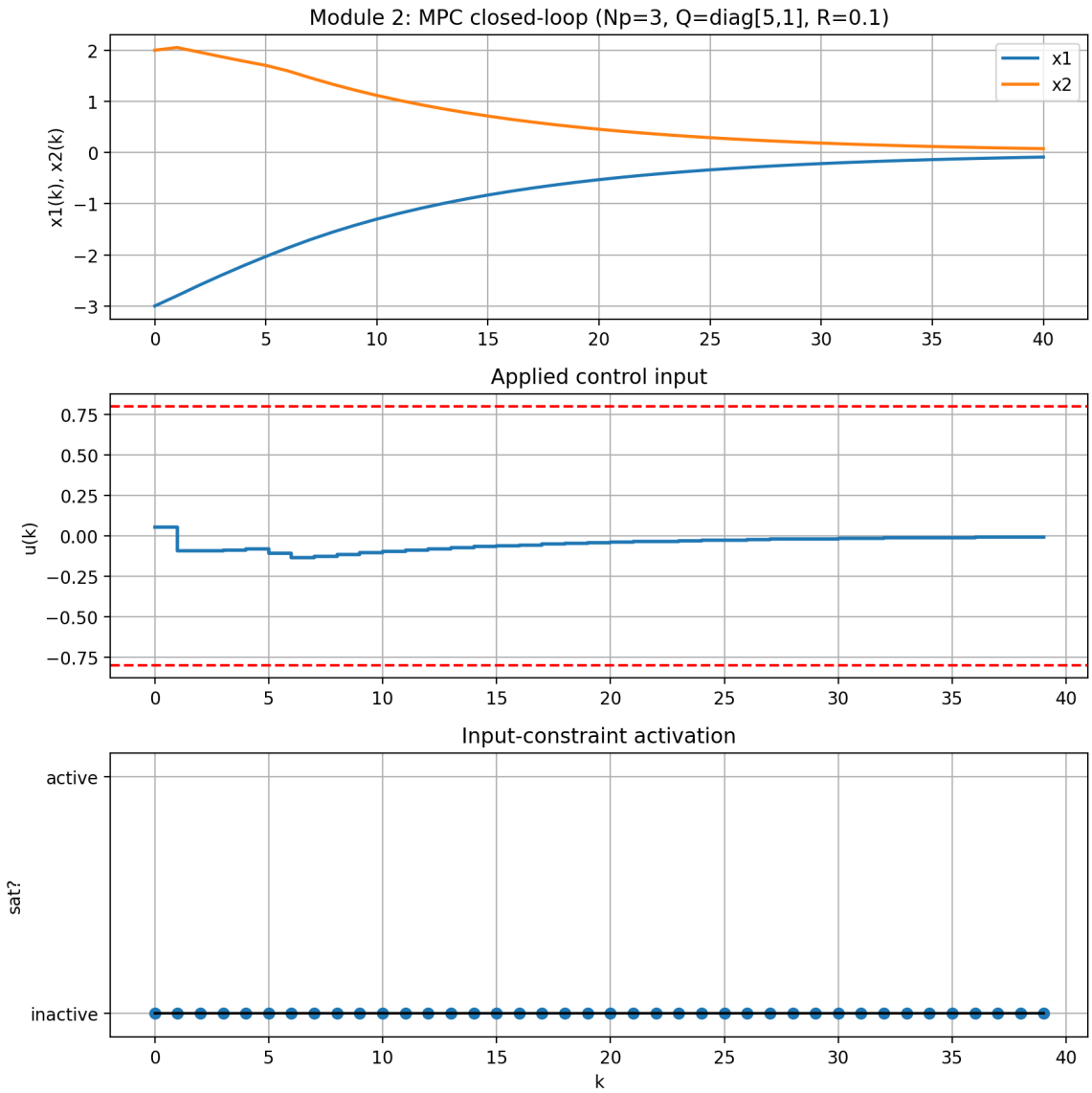

Figure 2: Module 2 constrained MPC: states, input, and input-constraint activity.

3.2 Analysis

The baseline run shows a conservative but stable controller response:

- states converge monotonically toward zero,
- the control sequence is smooth (small step-to-step variation),
- the input-constraint activity remains inactive.

This is consistent with the quadratic objective where state regulation is weighted more than velocity and input effort is still penalized.

What to add after extra runs. To complete the full assignment analysis, include additional runs for:

- horizon comparison ($N_p = 3, 10, 30$),
- cost tuning (increase Q, then increase R),
- terminal cost comparison (P from DARE).

Then discuss convergence speed, smoothness, and saturation differences.

4 Module 3: Tube MPC Intuition

4.1 Base Run (disturbance bound $= 0.2$)

Figure **??** compares the nominal and disturbed actual trajectories. Circular markers around the nominal path visualize the tube envelope used in the assignment intuition.

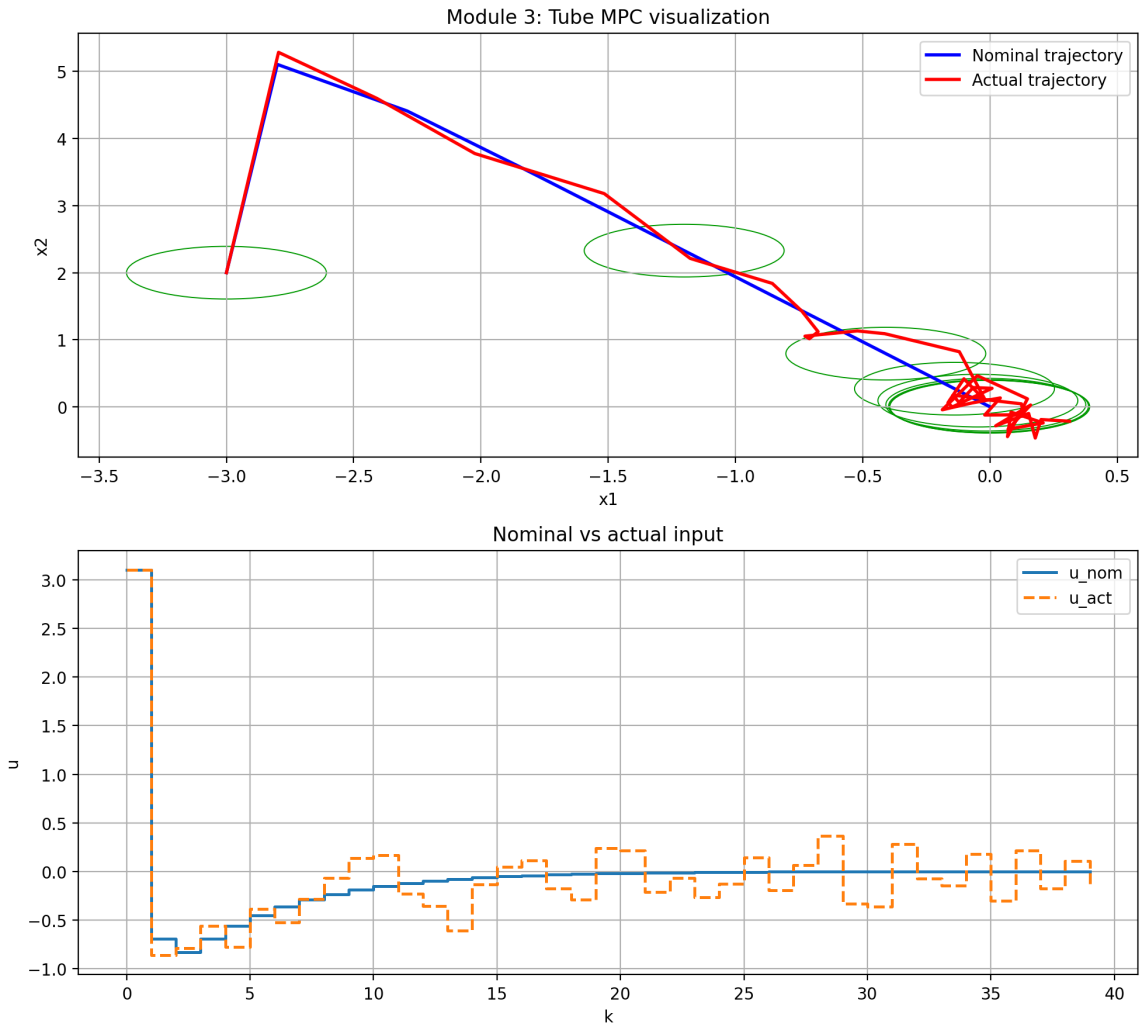

Figure 3: Module 3 tube MPC visualization: nominal vs actual trajectory and tube cross-sections.

4.2 Analysis

The ancillary feedback law

$$u_k = u_k^{\text{nom}} - Ke_k, \quad e_k = x_k - x_k^{\text{nom}}$$

keeps the disturbed trajectory near the nominal one despite random bounded disturbances. The resulting motion near the origin may still show a mild zig-zag pattern, which is expected from persistent disturbance injection even when $A - BK$ is Schur stable.

What to add after extra runs. For full coverage of the module tasks, add experiments with:

- multiple disturbance bounds (e.g., $0.1, 0.3, 0.5$),
- stronger vs weaker ancillary feedback (via $Q_{\text{error}}, R_{\text{error}}$).

Then compare tube width and control aggressiveness.

5 Conclusion

Across all modules, the simulations support the lecture concepts:

- open-loop prediction is useful but not robust to mismatch/disturbances,
- MPC provides structured closed-loop regulation with explicit cost/constraint trade-offs,
- tube-based feedback improves robustness by separating nominal planning from disturbance rejection.